Research Proposal

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Preliminary title:
Coercion and Hunger: District-Level Land Use Consolidation under Rwanda's Crop Intensification Program, Household Food Welfare, and Climate Vulnerability.

Research question:
How does district-level intensity of land use consolidation under Rwanda's Crop Intensification Program affect household food consumption welfare, and does this effect vary by household wealth?

Sub-questions:

- Is district-level LUC intensity negatively associated with household food consumption per adult equivalent?
- Does this effect increase as household wealth decreases?

Current knowledge:

Rwanda's Crop Intensification Program (CIP), launched in 2007 and ongoing, mandates smallholder farmers to cultivate state-assigned monocrops on consolidated plots, enforced through the *imihigo* performance contract system (Cioffo et al., 2016). The programme is widely cited as a success, producing nationally reported yield increases—though these have been contested on data quality grounds (Heinen, 2022). Existing research reveals a distributional cost: Del Prete et al. (2019), using EICV panel data from 2010/11 and 2013/14, find that land consolidation participation significantly reduced consumption of meat, fish, and fruit. Clay and Zimmerer (2020) argue qualitatively that CIP eroded the crop diversity and livelihood flexibility that previously buffered smallholders against climate shocks, concentrating vulnerability among the poorest. Their claim has not been tested quantitatively with recent nationally representative data. This matters beyond Rwanda: food insecurity driven by structural agricultural policy is an established pathway from climate stress to household-level conflict and instability (Hsiang et al., 2013). A policy that reduces food welfare while increasing climate exposure therefore carries implications not only for nutrition but for social stability.

The core argument is that *imihigo* enforcement removed the micro-level adaptive strategies smallholders use to manage risk—intercropping, crop diversification across moisture gradients, and flexible plot allocation— creating structural vulnerability (Clay and Zimmerer, 2020; Matin et al., 2018). CIP households locked into monocrops of government-selected crops, which Clay and Zimmerer show are among the riskiest in drought and heavy rain, have fewer options to absorb losses when shocks hit. This cost falls hardest on the poorest households, who additionally lack savings, credit access, and non-farm income as alternative buffers (Matin et al., 2018).

Hypotheses:

H1 Households in higher-LUC districts will have lower food consumption per adult equivalent than those in lower-LUC districts (*food welfare effect*).

H2 This effect will be larger for poorer households (*poverty-concentration hypothesis*).

Preliminary structure:

1. *From Policy Success to Distributional Question* — Introduction and relevance
2. *What the Literature Misses* — State of the art and the untested resilience claim
3. *Coercion, Specialisation, and Structural Vulnerability* — Theory and hypotheses
4. *Measuring the Cost* — Research design and operationalization
   - 4.1 Data and variables
   - 4.2 Methodology and results
5. *Who Bears the Burden?* — Conclusion and implications

Research design:

The study matches district-level LUC intensity—constructed as a population-weighted average share of agricultural land under consolidation from the NISR Seasonal Agricultural Survey (SAS) 2024 across all three seasons—to household-level data from the EICV7 (2023/24) survey ($n \approx 15{,}000$). The dependent variable is log food consumption per adult equivalent, drawn from the EICV7 poverty file. The analysis uses OLS regression with LUC intensity as the main independent variable, a LUC $\times$ wealth-quintile interaction term to test H2, and province fixed effects and urban/rural controls throughout. The climate and conflict relevance is addressed in the theoretical framework: CIP forces households into drought-vulnerable monocrops precisely in districts where agricultural climate shocks are most prevalent, and reduced food welfare is an established antecedent of household-level instability.

Data sources:

Variable	Source	Access
Food consumption per adult equivalent	EICV7 (2023/24) poverty file	World Bank Microdata Catalog; free registration
Household controls	EICV7 poverty file; savings & credit modules	Included in EICV7 package
LUC intensity	NISR Seasonal Agricultural Survey 2024, Seasons A, B & C	NISR microdata catalog; free registration

Literature:

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Cioffo, Giuseppe, An Ansoms, and Jude Murison. 2016. "Modernising Agriculture through a 'New' Green Revolution." *Review of African Political Economy* 43(148): 188–204.

Clay, Nathan, and Karl S. Zimmerer. 2020. "Who Is Resilient in Africa's Green Revolution?" *Land Use Policy* 97: 104558.

Del Prete, Davide, Luc Ghins, Emiliano Magrini, and Karl Pauw. 2019. "Land Consolidation, Specialization and Household Diets: Evidence from Rwanda." *Food Policy* 83: 139–149.

Heinen, Sebastian. 2022. "Rwanda's Agricultural Transformation Revisited." *Journal of Development Studies* 58(10): 2044–2064.

Hsiang, Solomon M., Marshall Burke, and Edward Miguel. 2013. "Quantifying the Influence of Climate on Human Conflict." *Science* 341(6151): 1235367.

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Matin, Nilufar, John Forrester, and Jonathan Ensor. 2018. "What Is Equitable Resilience?" *World Development* 109: 197–205.

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